University Of Engineering and Technology, Lahore Computer Engineering Department

Course Name: Programming Fundamentals and Data Structures	**Course Code: CMPE-112L**
Assignment Type: Lab Report	**Dated: 30th April, 2026**
Semester: 1st	**Session: 2026**
Lab/Project/Assignment #: 12	**CLOs to be covered: CLO2**
Lab Title: Lambda Functions	**Teacher Name: Hafiz Muhammad Mubashar**

Lab Evaluation:

CLO2	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	**Level2**	**Level3**	**Level4**	**Level5**	**Level6**
(5)						
					Total	/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 12: Lambda Functions

Lab Goals/Objectives:

To understand and use lambda (anonymous) functions in Python as a concise way to define small, single-expression functions.

Equipment Required:
Python 3, VS Code / PyCharm

Theory:
A lambda function is a small anonymous function defined using the lambda keyword. It can take any number of arguments but can only contain a single expression, which is implicitly returned.

Lab Work:

Task 1: Capital and Reverse

```
string= str(input("Enter sentence:"))
def  l_u(answer):
   return answer.upper()
lambda answer: answer.upper()

reverse_string = lambda r: r[::-1]

if str.islower(string):
   text= l_u(string)
   print(text)
elif str.isupper(string):
   print("Reverse:",reverse_string(string))
else:
   print("Please enter a valid sentence")
```

Result:

```
Enter sentence:hello my name is ann
HELLO MY NAME IS ANN
```

Task 2: Lambda - Square

```
a =lambda x: x**2
print(a(5))
```

Result:

```
25
```

Task 3: Lambda - Sum of Two
a = lambda x,y: x+y
print(a(3,6))

Result:

Task 4: Lambda - Sum of Three
a = lambda x,y,z: x+y+z
print(a(3,6,5))

Result:

Conclusions:

Lambda functions were used to write concise, single-expression functions, reinforcing an alternative to standard function definitions for simple operations.